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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Error in Measurements & Instruments

English Medium

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ERROR IN MEASUREMENTS & INSTRUMENTS

KEY CONCEPT

ERRORS

Whenever an experiment is performed, two kinds of errors can appear in the measured quantity.

(1) random and (2) systematic errors.

1. Random errors appear randomly because of operator, fluctuations in external conditions and variability of measuring instruments. The effect of random error can be some what reduced by taking the average of measured values. Random errors have no fixed sign or size.

2. Systematic errors occur due to error in the procedure, or miscalibration of the instrument etc. Such errors have same size and sign for all the measurements. Such errors can be determined.

A measurement with relatively small random error is said to have high precision. A measurement with small random error and small systematic error is said to have high accuracy.

The experimental error [uncertainty] can be expressed in several standard ways.

Error limits $Q \pm \Delta Q$ is the measured quantity and ΔQ is the magnitude of its limit of error. This expresses the experimenter's judgment that the 'true' value of Q lies between $Q - \Delta Q$ and $Q + \Delta Q$. This entire interval within which the measurement lies is called the range of error. Random errors are expressed in this form.

Absolute Error

Error may be expressed as absolute measures, giving the size of the error in a quantity in the same units as the quantity itself.

Least Count Error :– If the instrument has known least count, the absolute error is taken to be equal to the least count unless otherwise stated.

Relative (or Fractional) Error

Error may be expressed as relative measures, giving the ratio of the quantity's error to the quantity itself. In general,

$$\text{relative error} = \frac{\textit{absolute error in a measurement}}{\textit{size of the measurement}}$$

We should know the error in the measurement because these errors propagate through the calculations to produce errors in results.

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A. ***Systematic errors*** **:**

They have a known sign. The systematic error is removed before beginning calculations. Bench error and zero error are examples of systematic error.

B. ***Random error*** **:**

They have unknown sign. Thus they are represented in the form $A \pm a$.

Here we are only concerned with *limits of error*. We must assume a "worst-case" combination. In the case of substraction, A – B, the worst-case deviation of the answer occurs when the errors are either + a and – b or – a and + b. In either case, the maximum error will be (a + b).

For example in the experiment on finding the focal length of a convex lens, the object distance (u) is found by subtracting the positions of the object needle and the lens. If the optical bench has a least count of 1 mm, the error in each position will be 1 mm. So, the error in the value of u will be 2 mm.

1. ***Addition and subtraction rule*** **:** The absolute random errors add.

Thus if $R = A + B$, $r = a + b$ and if $R = A - B$, $r = a + b$

2. ***Product and quotient rule*** **:** The relative random errors add.

Thus if $R = AB$, $\frac{r}{R} = \frac{a}{A} + \frac{b}{B}$ and if $R = \frac{A}{B}$, then also $\frac{r}{R} = \frac{a}{A} + \frac{b}{B}$

3. ***Power rule*** **:** When a quantity Q is raised to a power P, the relative error in the result is P times the relative error in Q. This also holds for negative powers. If $R = Q^P$, $\frac{r}{R} = P \times \frac{q}{Q}$

4. The quotient rule is not applicable if the numerator and denominator are dependent on each other.

e.g if $R = \frac{XY}{X+Y}$. We cannot apply quotient rule to find the error in R. Instead we write the equation as follows $\frac{1}{R} = \frac{1}{X} + \frac{1}{Y}$. Differentiating both the sides, we get $-\frac{dR}{R^2} = -\frac{dX}{X^2} - \frac{dY}{Y^2}$.

Thus $\frac{r}{R^2} = \frac{x}{X^2} + \frac{y}{Y^2}$

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SIGNIFICANT DIGITS

Significant figures are digits that are statistically significant. There are two kinds of values in science:

1. Measured Values
2. Computed Values

The way that we identify the proper number of significant figures in science are different for these two types.

MEASURED VALUES

Identifying a measured value with the correct number of significant digits requires that the instrument's calibration be taken into consideration. The last significant digit in a measured value will be the first estimated position. For example, a metric ruler is calibrated with numbered calibrations equal to 1 cm. In addition, there will be ten unnumbered calibration marks between each numbered position. (each equal to 0.1 cm). Then one could with a little practice estimate between each of those marking. (each equal to 0.05 cm). That first estimated position would be the last significant digit reported in the measured value. Let's say that we were measuring the length of a tube, and it extended past the fourteenth numbered calibration half way between the third and fourth unnumbered mark. The metric ruler was a meter stick with 100 numbered calibrations. The reported measured length would be 14.35 cm. Here the total number of significant digits will be 4.

COMPUTED VALUE

The other type of value is a computed value. The proper number of significant figures that a computed value should have is decided by a set of conventional rules. However before we get to those rules for computed values we have to consider how to determine how many significant digits are indicated in the numbers being used in the math computation.

A. Rules for determining the number of significant digits in number with indicated decimals.

1. All nonzero digits (1-9) are to be counted as significant.
2. Zeros that have any nonzero digits anywhere to the LEFT of them are considered significant zeros.
3. All other zeros not covered in rule (2) above are NOT be considered significant digits.

For example : 0.0040000

The 4 is obviously to be counted significant (Rule-1), but what about the zeros? The first three zeros would not be considered significant since they have no nonzero digits anywhere to their left (Rule-3). The last four zeros would all be considered significant since each of them has the nonzero digit 4 to their left (Rule-2). Therefore the number has a total of five significant digits.

Here is another example : 120.00420

The digit 1, 2, 4 and 2 are all considered significant (Rule-1). All zeros are considered significant since they have non-zero digits somewhere to their left (Rule-2). So there are a total of eight significant digits. If in the question, we are given a number like 100, we will treat that the number has only one significant digit by convention.

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B. Determining the number of significant digits if number is not having an indicated decimal.

The decimal indicated in a number tells us to what position of estimation the number has been indicated. But what about 1,000,000?

Notice that there is no decimal indicated in the number. In other words, there is an ambiguity concerning the estimated position. This ambiguity can only be clarified by placing the number in exponential notation.

For example : If I write the number above in this manner.

1.00×10^6

I have indicated that the number has been recorded with three significant digits. On the other hand, if I write the same number as : 1.0000×10^6

I have identified the number to have 5 significant digits. Once the number has been expressed in exponential notation form then the digits that appear before the power of ten will all be considered significant. So for example : 2.0040×10^4 will have five significant digits. This means that unit conversion will not change the number of significant digits.

Thus 0.000010 km = 1.0 cm = 0.010 m = 1.0×10^{-2} m = 1.0×10^{-5} km

Rule for expressing proper number of significant digits in an answer from multiplication or division

For multiplication AND division there is the following rule for expressing a computed product or quotient with the proper number of significant digits.

The product or quotient will be reported as having as many significant digits as the number involved in the operation with the least number of significant digits.

For example : $0.000170 \times 100.40 = 0.017068$

The product could be expressed with no more that three significant digits since 0.000170 has only three significant digits, and 100.40 has five. So according to the rule the product answer could only be expressed with three significant digits. Thus the answer should be 0.0171 (after rounding off)

Another example : $2.000 \times 10^4 / 6.0 \times 10^{-3} = 0.33 \times 10^7$

The answer could be expressed with no more that two significant digits since the least digit number involved in the operation has two significant digits.

Sometimes this would required expressing the answer in exponential notation.

For example : $3.0 \times 800.0 = 2.4 \times 10^3$

The number 3.0 has two significant digits and then number 800.0 has four. The rule states that the answer can have no more than two digits expressed. However the answer as we can all see would be 2400. How do we express the answer 2400 while obeying the rules? The only way is to express the answer in exponential notation so 2400 could be expressed as : 2.4×10^3

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Rule for expressing the correct number of significant digits in an addition or substraction :

The rule for expressing a sum or difference is considerably different than the one for multiplication of division. The sum or difference can be no more precise than the least precise number involved in the mathematical operation. Precision has to do with the number of positions to the RIGHT of the decimal. The more position to the right of the decimal, the more precise the number. So a sum or difference can have no more indicated positions to the right of the decimal as the number involved in the operation with the LEAST indicated positions to the right of its decimal.

For example : 160.45 + 6.732 = 167.18 (after rounding off)

The answer could be expressed only to two positions to the right of the decimal, since 160.45 is the least precise.

Another example : 45.621 + 4.3 – 6.41 = 43.5 (after rounding off)

The answer could be expressed only to one position to the right of the decimal, since the number 4.3 is the least precise number (i.e. having only one position to the right of its decimal). Notice we aren't really determining the total number of significant digits in the answer with this rule.

Rules for rounding off digits :

There are a set of conventional rules for rounding off.

1. Determine according to the rule what the last reported digit should be.
2. Consider the digit to the right of the last reported digit.
3. If the digit to the right of the last reported digit is less than 5 round it and all digits to its right off.
4. If the digit to the right of the last reported digit is greater than 5 round it and all digits to its right off and increased the last reported digit by one.
5. If the digit to the right of the last reported digit is a 5 followed by either no other digits or all zeros, round it and all digits to its right off and if the last reported digit is odd round up to the next even digit. If the last reported digit is even then leave it as is.

   For example if we wish to round off the following number to 3 significant digits : 18.3682

   The last reported digits would be the 3. The digit to its right is a 6 which is greater than 5. According to the Rule-4 above, the digit 3 is increased by one and the answer is : 18.4

   Another example : Round off 4.565 to three significant digits.

   The last reported digit would be the 6. The digit to the right is a 5 followed by nothing. Therefore according to Rule-5 above since the 6 is even it remains so and the answer would be 4.56.

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MEASUREMENT OF LENGTH USING VERNIER CALLIPERS & SCREW GAUGE

The simplest method measuring the length of a straight line is by means of a meter scale. But there exists some limitation in the accuracy of the result:

(i) the dividing lines have a finite thickness.

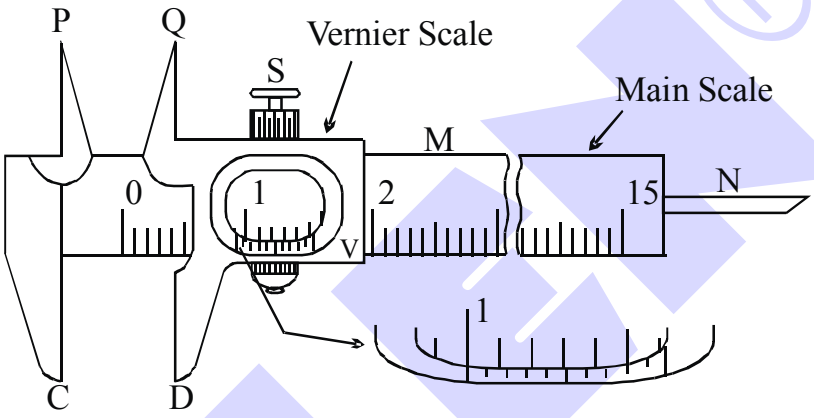
(ii) naked eye cannot correctly estimate less than 0.5 mm

For greater accuracy devices like

(a) Vernier callipers

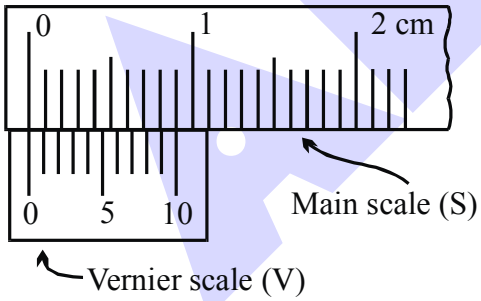
(b) micrometer scales (screw gauge) are used.

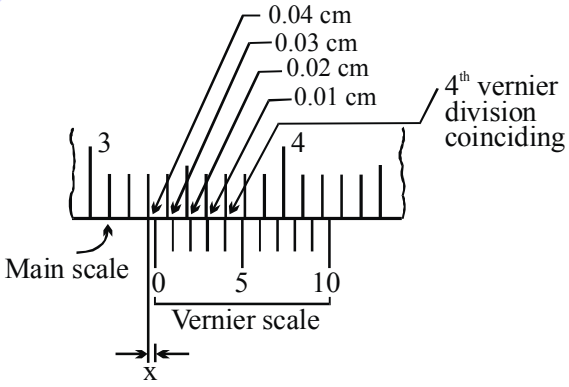
- ***VERNIER CALLIPERS***



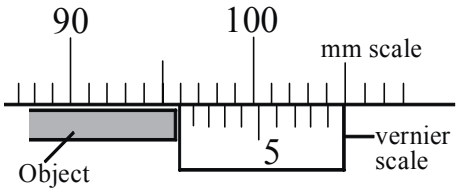
Vernier Callipers

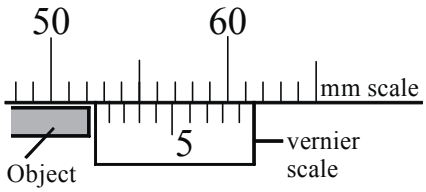
Principle of Vernier





Reading a vernier with 4th division coinciding





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It consists of a main scale graduated in cm/mm over which an auxiliary scale (or Vernier scale) can slide along the length. The division of the Vernier scale being shorter than the divisions of the main scale.

Least count of Vernier Callipers

The least count or Vernier constant (v. c) is the minimum value of correct estimation of length without eye estimation. If N division of vernier coincides with (N-1) division of main scale, then

$$N\,(VS) = (N-1)\text{ ms} \quad \Rightarrow \quad 1VS = \frac{N-1}{N}\text{ ms}$$

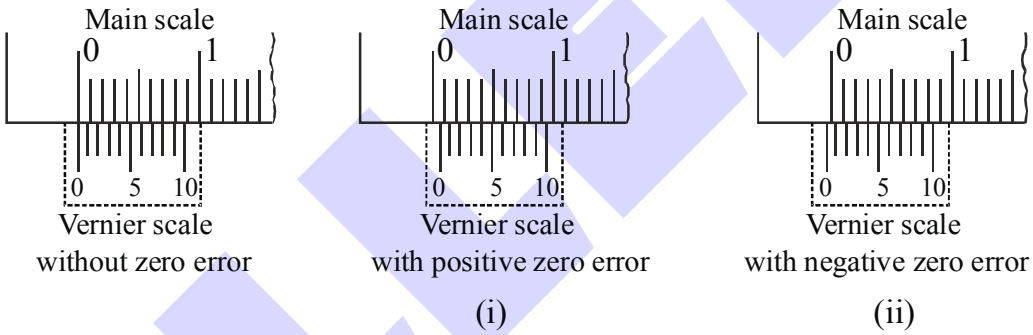
Vernier constant = 1 ms – 1 vs = $\left(1-\frac{N-1}{N}\right)$ms = $\frac{1\text{ms}}{N}$, which is equal to the value of the smallest division on the main scale divided by total number of divisions on the vernier scale.

Length as measured by Vernier Callipers

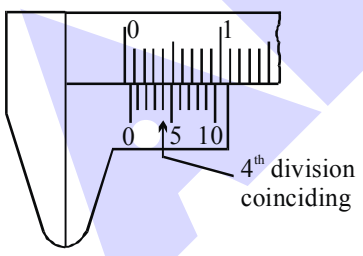
The formula for measuring the length is L = main scale reading + least count of vernier scale × Vernier scale division coinciding with a main scale division

Main scale reading is given by the zeroth division of the vernier scale as shown in the figure.

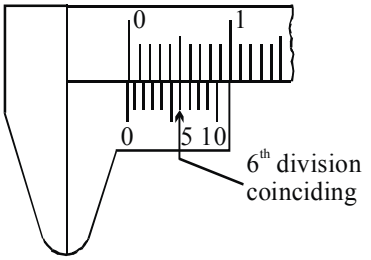
Zero error:



(i) (ii)



Positive zero error (+0.04 cm)
and its correction



Negative zero error = (–0.04 cm)
and its correction

If the zero marking of main scale and vernier callipers do not coincide, necessary correction has to be made for this error which is known as zero error of the instrument.

If the zero of the vernier scale is to the right of the zero of the main scale the zero error is said to be positive and the correction will be negative and vice versa.

The zero error is always subtracted from the reading to get the corrected value.

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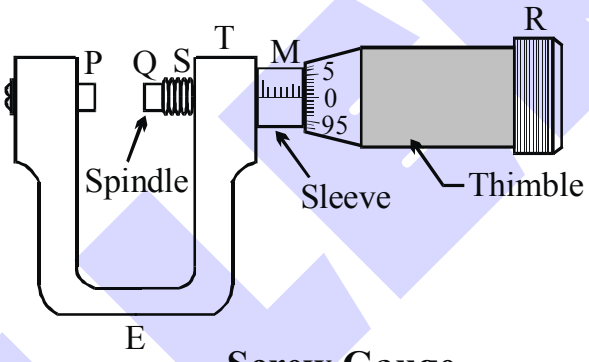
If the zero error is positive, its value is calculated as we take any normal reading. If the zero error negative (the zero of vernier scale lies to the left of the zero of main scale),

negative zero error = – [Total no. of vsd – vsd coinciding] × L.C.

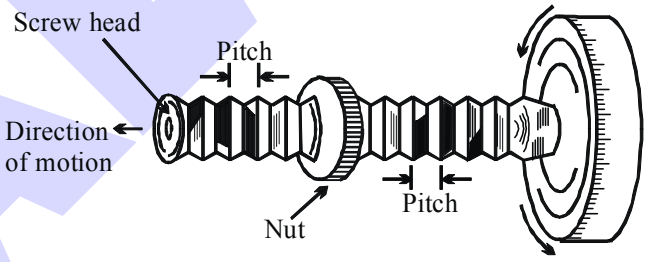
Do not try to read the main scale at the point where the lines match best. This has no meaning. Read from the vernier scale instead. Sometimes it is difficult to tell whether the best match of lines is for vernier marks 9, 0, or 1. Make your best estimate, but realize that the final result including the vernier must round off to the result you would choose if there was no vernier. If the mark is close to 3.20 on the main scale, but the vernier reading is 9, the length is 3.19 cm. If the mark is close to 3.2 on the main scale and the vernier is 1, the length is 3.21 cm.

- ***SCREW GAUGE (OR MICROMETER SCREW)***

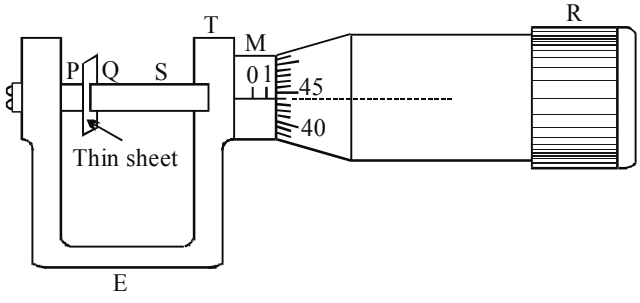
In general vernier callipers can measure accurately upto 0.01 em and for greater accuracy micrometer screw devices e.g. screw gauge, spherometer are used. These consist of accurately cut screw which can be moved in a closely fitting fixed nut by tuming it axially. The instrument is provided with two scales:



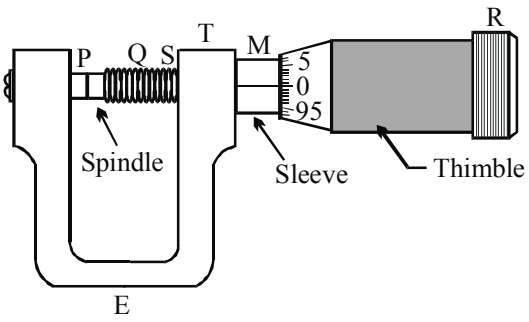
Screw Gauge



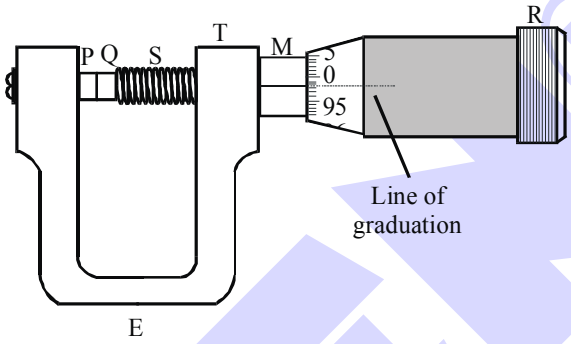
Principle of a micrometer



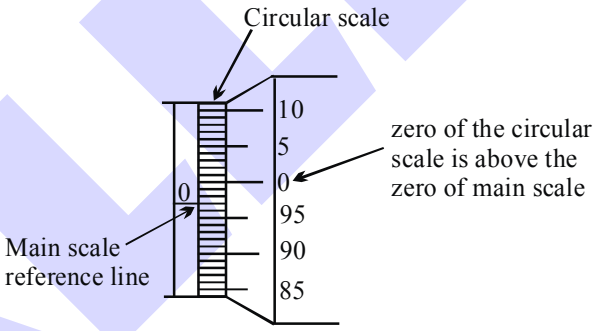
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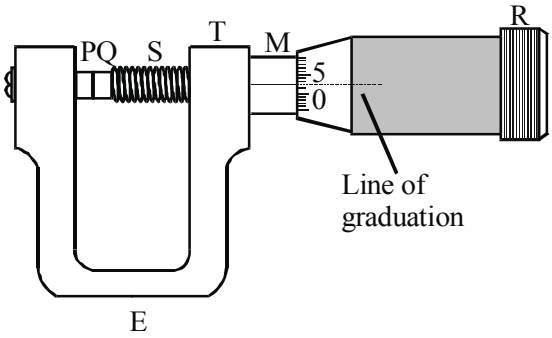


Screw gauge with no zero error



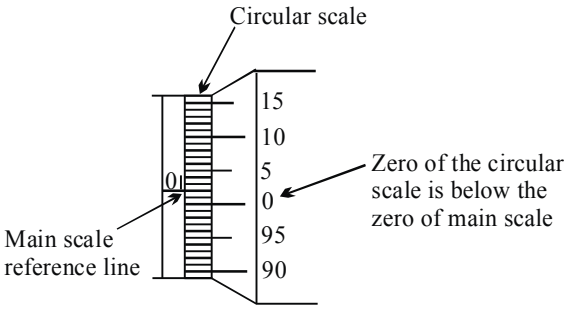
Negative zero error
(3 division error) i.e., - 0.003 cm





Positive zero error
(2 division error) i.e., + 0.002 cm

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(i) The main scale or pitch scale M graduated along the axis of the screw.
(ii) The cap-scale or head scale H round the edge of the screw head.

Constants of the Screw Gauge

(a) Pitch : The translational motion of the screw is directly proportional to the total rotation of the head. The pitch of the instrument is the distance between two consecutive threads of the screw which is equal to the distance moved by the screw due to one complete rotation of the cap. Thus for 10 rotation of cap = 5 mm, then pitch = 0.5 mm

(b) Least count : In this case also, the minimum (or least) measurement (or count) of length is equal to one division on the head scale which is equal to pitch divided by the total cap divisions. Thus in the aforesaid Illustration:, if the total cap division is 100, then least count = 0.5mm/100 = 0.005 mm

(c) Measurement of length by screw gauge :

L = n × pitch + f × least count, where n = main scale reading & f = caps scale reading

Zero Error : In a perfect instrument the zero of the head scale coincides with the line of graduation along the screw axis with no zero-error, otherwise the instrument is said to have zero-error which is equal to the cap reading with the gap closed. This error is positive when zero line or reference line of the cap lies **below** the line of graduation and versa. The corresponding corrections will be just opposite.

Backlash Error

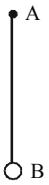
In a good instrument (either screw gauge or a spherometer) the thread on the screw and that on the nut (in which the screw moves), should tightly fit with each other. However, with repeated use, the threads of both the screw and the nut may get worn out. As a result a gap develops between these two threads, which is called "play". The play in the threads may introduce an error in measurement in devices like screw gauge. This error is called backlash error. In instruments having backlash error, the screw slips a small linear distance without rotation. To prevent this, it is advised that the screw should be moved in only one direction while taking measurements.

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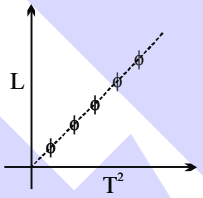
EXPERIMENTS IN JEE-ADVANCE SYLLABUS

1. MEASUREMENT OF g USING A SIMPLE PENDULUM

A small spherical bob is attached to a cotton thread and the combination is suspended from a point A. The length of the thread (L) is read off on a meter scale. A correction is added to L to include the finite size of the bob and the hook. The corrected value of L is used for further calculation. The bob is displaced slightly to one side and is allowed to oscillate, and the total time taken for 50 complete oscillations is noted on a stop-watch.



The time period (T) of a single oscillation is now calculated by division. Observations are now taken by using different lengths for the cotton thread (L) and pairs of values of L and T are taken. A plot of L v/s T^2, on a graph, is linear. g is given by $g = 4\pi^2 \frac{L}{T^2}$



The major errors in this experiment are

(a) Systematic : Error due to finite amplitude of the pendulum (as the motion is not exactly SHM). This may be corrected for by using the correct numerical estimate for the time period. However the practice is to ensure that the amplitude is small.

(b) Statistical : Errors arising from measurement of length and time. $\frac{\delta g}{g} = \frac{\delta L}{L} + 2\left(\frac{\delta T}{T}\right)$

The contributions to δL, δT are both statistical and systematic. These are reduced by the process of averaging. The systematic error in L can be reduced by plotting several values of L vs T^2 and fitting to a straight line. The slope of this fit gives the correct value of L/T^2

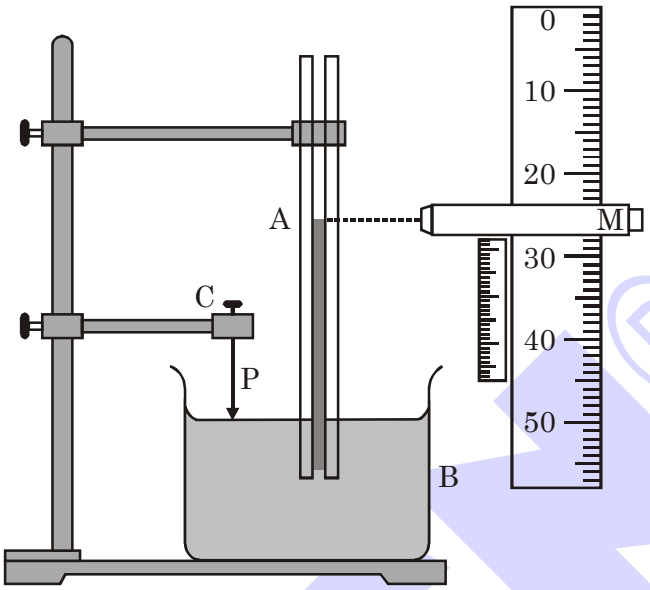
2. MEASURMENT OF SURFACE TENSION OF WATER

AIM : To determine the surface tension of water by capillary rise method.

Apparatus and Material Required : A glass/plastic capillary tube, travelling microscope, beaker, cork with pin, clamps and stand, thermometer, dilute nitric acid solution, dilute caustic soda solution, water, plumb line.

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Principle : When a liquid rises in a capillary tube [Fig. E 12.1], the weight of the column of the liquid of density ρ below the meniscus, is supported by the upward force of surface tension acting around the circumference of the points of contact. Therefore



$2\pi rT = \pi r^2 h\rho g$ (approx) for water

or $$T = \frac{h\rho gr}{2}$$

where T = surface tension of the liquid,
h = height of the liquid column and
r = inner radius of the capillary tube

Procedure :

1. Do the experiment in a well-lit place for example, near a window or use an incandescent bulb.
2. Clean the capillary tube and beaker successively in caustic soda and nitric acid and finally rinse thoroughly with water.
3. Fill the beaker with water and measure its temperature.
4. Clamp the capillary tube near its upper end, keeping it above the beaker. Set it vertical with the help of a plumbline held near it. Move down the tube so that its lower end dips into the water in the beaker.
5. Push a pin P through a cork C, and fix it on another clamp such that the tip of the pin is just above the water surface as shown in figure. Ensure that the pin does not touch the capillary tube. Slowly lower the pin till its tip just touches the water surface. This can be done by coinciding the tip of the pin with its image in water.
6. Now focus the travelling microscope M on the meniscus of the water in capillary A, and move the microscope until the horizontal crosswire is tangential to the lowest point of the meniscus, which is seen inverted in M. If there is any difficulty in focussing the meniscus, hold a piece of paper at the lowest point of the meniscus outside the capillary tube and focus it first, as a guide. Note the reading of travelling microscope.

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7. Mark the position of the meniscus on the capillary with a pen. Now carefully remove the capillary tube from the beaker, and then the beaker without disturbing the pin.
8. Focus the microscope on the tip of the pin and note the microscope reading.
9. Cut the capillary tube carefully at the point marked on it. Fix the capillary tube horizontally on a stand. Focus the microscope on the transverse cross section of the tube and take readings to measure the internal diameter of the tube in two mutually perpendicular directions.

Observations :

Determination of h

Least count (L.C.) of the microscope = ... mm

Table E : Measurement of capillary rise

S.no.	Reading of meniscus h_1 (cm)			Reading of tip of pin touchng surface of water h_2 (cm)			$h = h_1 - h_2$
	M.S.R. S (cm)	V.S.R. n	$h_1 = (S + n \times L.C.)$	M.S.R. S' (cm)	V.S.R. n'	$h_2 = (S' + n' \times L.C.)$ (cm)	
1							
2							
3							

Table E : Measurement of diameter of the capillary tube

S.no.	Reading along a diameter (cm)		Diameter $d_1 (x_2 - x_1)$	Reading along perpendicular diameter		Diameter $d_2 (y_2 - y_1)$	Mean diameter d
	One end	Other end	(cm)	One end	Other end	(cm)	$= \frac{d_1 + d_2}{2}$
1	x_1	x_2		y_1	y_2		
2							
3							

Precautions :

1. To make capillary tube free of contamination, it must be rinsed first in a solution of caustic soda then with dilute nitric acid and finally cleaned with water thoroughly.
2. The capillary tube must be kept vertical while dipping it in water.
3. To ensure that capillary tube is sufficiently wet, raise and lower water level in container by lifting or lowering the beaker. It should have no effect on height of liquid level in the capillary tube.
4. Water level in the capillary tube should be slightly above the edge of the beaker/dish so that the edge does not obstruct observations.
5. Temperature should be recorded before and after the experiment.
6. Height of liquid column should be measured from lowest point of concave meniscus.

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Sources of Error :

1. Inserting dry capillary tube in the liquid can cause gross error in the measurement of surface tension as liquid level in capillary tube may not fall back when the level in container is lowered.
2. Surface tension changes with impurities and temperature of the liquid.
3. Non-vertical placement of the capillary tube may introduce error in the measurement of height of the liquid column in the tube.
4. Improper focussing of meniscus in microscope could cause an error in measurement of the height of liquid column in the capillary tube.

Discussion :

1. In a fine capillary tube, the meniscus surface may be considered to be hemispherical and the weight of the liquid above the lowest point of the meniscus as $\frac{1}{3}\rho r^3 \pi g$. Taking this into account, the formula for surface tension is modified to $T = \frac{1}{2}\rho g r\left(h + \frac{r}{3}\right)$. More precise calculation of surface tension can be done using this formula.
2. If the capillary is dry from inside the water that rises to a certain height in it will not fall back, so the capillary should be wet from inside. To wet the inside of the capillary tube thoroughly, it is first dipped well down in the water and raised and clamped. Alternatively, the beaker may be lifted up and placed down.

3. DETERMINATION OF THE SPECIFIC HEAT CAPACITY OF A GIVEN LIQUID BY THE METHOD OF MIXTURES.

Apparatus and material required :

Copper calorimeter with lid, stirrer and insulating cover (the lid should have provision to insert thermometer in addition to the stirrer), two thermometers (0 °C to 100 °C or 110 °C with a least count of 0.5 °C), a solid, preferably metallic (brass/copper/steel/aluminium) cylinder which is insoluble in given liquid and water, given liquid, two beakers (100 mL and 250 mL), a heating device (heater/hot plate/gas burner); physical balance, spring balance with weight box (including fractional weights), a piece of strong nonflexible thread (25-30 cm long), water, laboratory stand, tripod stand and wire gauze.

Principle/Theory :

For a body of mass m and specific heat s, the amount of heat Q lost/gained by it when its temperature falls/rises by Δt is given by

$$\Delta Q = ms\,\Delta t$$

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Specific heat capacity: It is the amount of heat required to raise the temperature of unit mass of a substance through 1°C. Its S.I unit is $Jkg^{-1} K^{-1}$.

Principle of Calorimetry: If bodies of different temperatures are brought in thermal contact, the amount of heat lost by the body at higher temperature is equal to the amount of heat gained by the body at lower temperature, at thermal equilibrium, provided no heat is lost to the surrounding.

Procedure :

1. Set the physical balance and make sure there is no zero error.
2. Weigh the empty calorimeter with stirrer and lid with the physical balance/spring balance. Ensure that calorimeter is clean and dry. Note the mass m1 of the calorimeter. Pour the given liquid in the calorimeter. Make sure that the quantity of liquid taken would be sufficient to completely submerge the solid in it. Weigh the calorimeter with liquid along with the stirrer and the lid and note its mass m_2. Place the calorimeter in its insulating cover.
3. Take a metallic cylinder whose specific heat capacity is known. Dip it in water in a container and shake it to remove the water sticking to its surface. Weigh the wet solid with the physical balance and note down its mass m_3.
4. Tie the solid tightly with the thread at its middle. Make sure that it can be lifted by holding the thread without slipping.
   Place a 250 mL beaker on the wire gauze kept on a tripod stand as shown in figure. Fill the beaker up to half with water. Now suspend the solid in the beaker containing water by tying the other end of the thread to a laboratory stand. The solid should be completely submerged in water and should be atleast 0.5 cm below the surface. Now heat the water with the solid suspended in it [figure].
5. Note the least count of the thermometer. Measure the temperature of the water taken in the calorimeter. Record the temperature t_1 of the water.
6. Let the liquid in the beaker boil for about 5-10 minutes. Now measure the temperature t_2 of the liquid with the other thermometer and record the same. Holding the solid with the thread tied to it remove it from the boiling water, shake it to remove water sticking on it and quickly put it in the liquid in the calorimeter and replace the lid immediately [Figure]. Stir it with the stirrer. Measure the temperature of the liquid once equilibrium is attained, that is, temperature of the mixture becomes constant. Record this temperature as t_3.

Observations :

Mass of the empty calorimeter with stirrer (m_1)	= ... g
Mass of the calorimeter with liquid (m_2)	= ... g
Mass of solid (m_3)	= ... g
Initial temperature of the liquid (t_1)	= °C = ... K
Temperature of the solid in boiling water (t_2)	= °C = ... K
Temperature of the mixture (t_3)	= °C = ... K
Specific heat capacity of material of calorimeter s_1	= ... Jkg^{-1} °C^{-1} ($Jkg^{-1} K^{-1}$)
Specific heat capacity of solid (s_0)	= ... $Jkg^{-1} K^{-1}$

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Calculations :

1. Mass of the liquid in calorimeter $(m_2 - m_1)$ = ... g = ... kg
2. Change in temperature of liquid and calorimeter $(t_3 - t_1)$ = ... °C
3. Change in temperature of solid $(t_2 - t_3)$ = ... °C
   Heat given by solid in cooling from t_2 to t_3.
   = Heat gained by liquid in raising its temperature from t_1 to t_3 +
   heat gained by calorimeter in raising its temperature from t_1 to t_3.
   $m_3 s_o (t_2 - t_3) = (m_2 - m_1)\, s\, (t_2 - t_1) + m_1 s_1 (t_3 - t_1)$

$$s = \frac{m_3 s_0 (t_2 - t_3) - m_1 s_1 (t_3 - t_1)}{(m_2 - m_1)(t_2 - t_1)} = \text{... J kg}^{-1}\,^\circ\text{C}^{-1}$$

Precautions :

1. Physical balance should be in proper working condition and ensure that there is no zero error.
2. The two thermometers used should be of the same range and least count.
3. The solid used should not be chemically reactive with the liquid used or water.
4. The calorimeter should always be kept in its insulated cover and at a sufficient distance from the source of heat and should not be exposed to sunlight so that it absorbs no heat from the surrounding.
5. The solid should be transferred quickly so that its temperature is same as recorded when it is dropped in the liquid.
6. Liquid should not be allowed to splash while dropping the solid in it in the calorimeter. It is advised that the solid should be lowered gently into the liquid with the help of the thread tied to it.
7. While measuring the temperature, the thermometers should always be held in vertical position. The line of sight should be perpendicular to the mercury level while recording the temperature.

Sources of error :

1. Radiation losses cannot be completely eliminated.
2. Heat loss that takes place during the short period while transferring hot solid into calorimeter, cannot be accounted for.
3. Though mercury in the thermometer bulb has low specific heat, it absorbs some heat.
4. There may be some error in measurement of mass and temperature.

4. DETERMINATION OF REFRACTIVE INDEX OF A GLASS SLAB USING A TRAVELLING MICROSCOPE.

Apparatus and material required :

A travelling microscope, a glass slab, lycopodium powder/chalk dust and a paper.

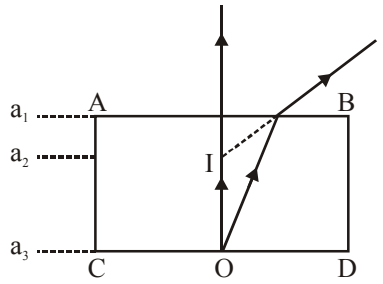


Figure : Formation of image I of a point O in a glass slab

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Principle :

If a glass slab is placed in air on a horizontal surface and its bottom surface is viewed from top, it appears to be elevated due to the phenomenon of refraction. The distance between this apparent bottom and the top surface of the slab gives the apparent thickness of the slab. In case of normal observation, it can be shown that the refractive index of glass with respect to the medium, air is,

$$n_{ga} = \frac{\text{real thickness of the slab}}{\text{apparent thickness of the slab}}$$

Procedure :

1. Find the least count of the microscope scale, being used.
2. Put a mark on a sheet of paper.
3. Place the paper on the horizontal platform of the microscope. Adjust the microscope in such a way that its lens system is vertically above the mark.
4. Focus the microscope on the mark and record the reading a_1 using Main Scale Reading (MSR) and Vernier Scale Reading (VSR) of its coinciding division as shown in the table.
5. Next, place the glass slab over the mark on the sheet of paper.
6. Move the microscope upward until mark on the paper, seen through the slab is sharp and clear. Take the reading a_2 with the main scale and coinciding division of the vernier scale.
7. Spread a little lycopodium powder/chalk dust over the top surface of the glass slab.
8. Raise the lens system of the microscope and focus the microscope to see some of their particles clearly and record reading a_3.
9. Invert the slab and repeat steps 3 to 8.

Observations :

Least Count (LC) of the travelling microscope :

20 Main Scale divisions (MSD) = 1 cm (say)

$\therefore$ 1 MSD = $\frac{1}{20}$cm

50 Vernier Scale Divisons (VSD) = 49 MSD (say)

$\therefore$ $1\text{VSD} = \frac{49}{50}\text{MSD} = \frac{49}{50} \times \frac{1}{20}\text{cm}$

Least Count of the microscope = (1 MSD – 1 VSD)

$= \frac{1}{20} - \frac{49}{50} \times \frac{1}{20} = \frac{1}{20}\ 1 - \frac{49}{50}\text{cm}$

LC = 0.001 cm

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Table : Refractive index of glass slab

S. no.	Reading of the microscope when focused on								
	Mark made on paper			Mark on paper through the slab			Particles on top of the glass surface		
	M.S.R. M (cm)	V.S.R. N (cm)	$a_1 = M + N \times L.C.$ (cm)	M.S.R. M (cm)	V.S.R. N (cm)	$a_2 = M + N \times L.C.$ (cm)	M.S.R. M (cm)	V.S.R. N (cm)	$a_3 = M + N \times L.C.$ (cm)
1									
2									
3									

Calculations :

The refractive index n_{ga} is given by

$$n_{ga} = \frac{\text{real thickness of the slab}}{\text{apparent thickness of the slab}} = \frac{a_1 - a_3}{a_1 - a_2} = \frac{b}{c} \text{ (say)}$$

Find n_{ga} for the two sets of observations and then the mean value of n_{ga}.

Error

The estimated uncertainty in the measuremtn of n_{ga} is

$$\frac{\Delta n_{ga}}{n_{ga}} = \frac{\Delta b}{b} + \frac{\Delta c}{c} \quad \text{... (i)}$$

where $$\frac{\Delta b}{b} = \left(\frac{\Delta a_1}{a_1} + \frac{\Delta a_3}{a_3}\right) \quad \text{... (ii)}$$

and $$\frac{\Delta c}{c} = \left[\frac{\Delta a_1}{a_1} + \frac{\Delta a_2}{a_2}\right] \quad \text{... (iii)}$$

From equation (i)

$$\frac{\Delta n_{ga}}{n_{ga}} = \left(\frac{2\Delta a}{b} + \frac{2\Delta a}{c}\right)$$

or $$\Delta n_{ga} = 2n_{ga}\left[\frac{\Delta a}{b} + \frac{\Delta a}{c}\right]$$

Δa = least count of microscope and hence uncertainty in a_1, a_2 and a_3 measurements.

Maximum value of Δn_{ga} as obtained from the two sets of observations should be reported with the result as experimental error.

Precautions :

1. The screws used in focussing the microscope should be moved in only one direction to aboid back-lash error.
2. Once the microscope is focused for the first reading i.e., a_1, the focusing arrangement in the lens system should not be changed/altered for subsequent readings namely for a_2 and a_3.

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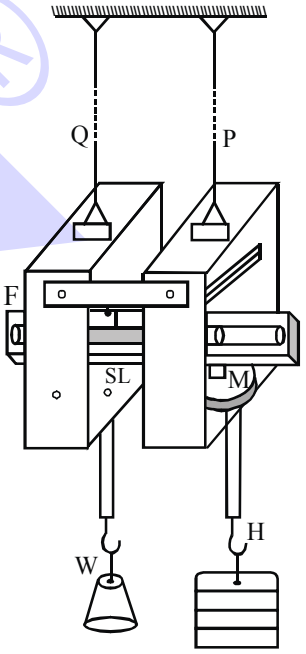
3. The glass slab should be placed on a horizontal surface.
4. Use hand lens/magnifying glass to read the vernier scale to avoid error in finding vernier coinciding divsion.

Sources of error :

1. Position of the microscope may not be normal to the surface of the glass slab.
2. If the layer of lycopodium powder/chalk dust spread on the glass slab is thick, it will not actually represent the top of the glass slab and thus produce error in the result.

5. DETERMINATION OF YOUNG'S MODULUS BY SEARLE'S METHOD

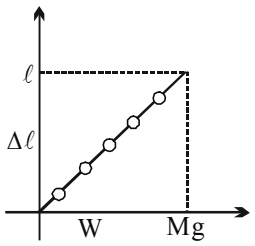
The experimental set up consists of two identical wires P and Q of uniform cross section suspended from a fixed rigid support. The free ends of these parallel wires are connected to a frame F as shown in the figure. The length of the wire Q remains fixed while the load L attached to the wire P through the frame F is varied in equal steps so as to produce extension along .the length. The extension thus produced is measured with the help of spirit level SL and micrometer screw M attached to the F frame on the side of the experimental wire. On placing the slotted weights on the hanger H upto a permissible value (half of the breaking force) the wire gets extended by small amount and the spirit level gets disturbed from horizontal setting. This increase in length is measured by turning the micrometer screw M upwards so as to restore the balance of the spirit level. If n be the number of turns of the micrometer screw and f be the difference in the cap reading, the increase in length M is obtained by Δl = n × pitch + f × least count



In some situations, the change in length is obtained by vernier arrangement instead of the screw gauge.

The load on the hanger is reduced in the same steps and spirit level is restored to horizontal position. The mean of these two observations gives the true increase in length of the wire corresponding to the given value of load. This is to eliminate the effect of hysteresis.

From the data obtained, a graph showing extension (Δl) against the load (W) is plotted which is obtained as a straight line passing through the origin.



The slope of the line gives $\tan\theta = \frac{l}{W} = \frac{l}{Mg}$

Now, stress = $\frac{Mg}{\pi r^2}$ and strain = $\frac{l}{L}$

Y = Stress/ strain = $\frac{MgL}{\pi r^2 l} = \frac{L}{\pi r^2 \tan\theta}$

With known values of initial length L, radius r of the experimental wire and tanθ, Young's modulus Y can be calculated.

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6. FOCAL LENGTH OF A CONCAVE MIRROR AND A CONVEX LENS USING THE U-V METHOD.

In this method one uses an optical bench and the convex lens (or the concave mirror) is placed on the holder.

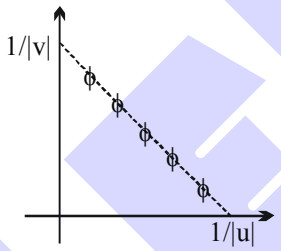
The position of the lens is noted by reading the scale at the bottom of the holder. A bright object (a filament lamp or some similar object) is placed at a fixed distance (u) in front of the lens (mirror).

The position of the image (v) is determined by moving a white screen behind the lens until a sharp image is obtained (for real images).

For the concave mirror, the position of the image is determined by placing a sharp object (a pin) on the optical bench such that the parallax between the object pin and the image is nil.

A plot of |u| versus |v| gives a rectangular hyperbola. A plot of $\frac{1}{|v|}$ vs $\frac{1}{|u|}$ gives a straight line.

The intercepts are equal to $\frac{1}{|f|}$, where *f* is the focal length.



Error : The systematic error in this experiment is mostly due to improper position of the object on the holder. This error maybe eliminated by reversing the holder (rotating the holder by 180° about the vertical) and then taking the readings again. Averages are then taken.

The equation for random errors gives: $\frac{\delta f}{f^2} = \frac{\delta u}{u^2} + \frac{\delta v}{v^2}$

The errors δu, δv correspond to the error in the measurement of u and v. Actually, we know the errors in the object position, lens position & image position. So, the errors in u & v are too be estimated as described before.

Index Error or Bench Error and its correction:

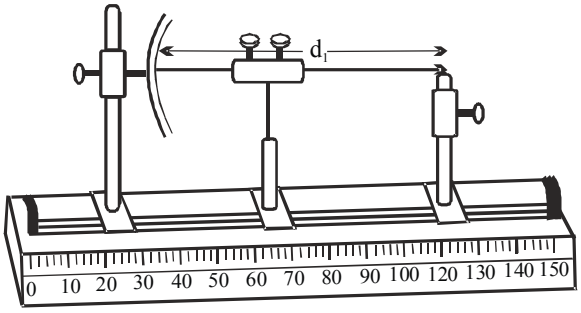
In an experiment using an optical bench we are required to measure the object and image distances from the pole or vertex on the mirror. The distance between the tip of the needles and the pole of the mirror is the actual distance. But we practically measure distances between the indices with the help of the scale engraved on the bench. These distances are called the observed distances. The actual distances may not be equal to the observed distances and due to this reason an error creeps in the measurement of the distances. This error is called the index or the bench error. This error is estimated with the help of a needle of known length placed horizontally between the tip of the needle and the pole.

Index Error = Observed distance – actual distance and

Index Correction = Actual – observed distance

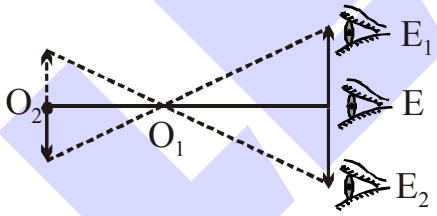
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Note: Index correction whether positive or negative, is always added algebraically to the observed distance to get the corrected distance.



Parallax

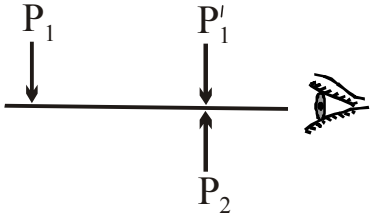
When two objects O_1 and O_2 are placed in such a way that both of them lie in the same line of sight as shown in figure, then the object nearer to the eye covers the object farther from it. Their images on the retina are superimposed and therefore, it is impossible to decide which is the nearer object. To identify this fact, the observer displaces his eye to a position E_1 or E_2 until he is able to see two distinct objects.



The more distant object O_2 apparently moves in the direction opposite to the displacement of the observer's eye with respect to the nearer object O_1. This relative shift in the position of two objects due to the shift in the position of the observer's eye is called parallax.

Parallax between the two objects disappear if they are at the same position.

The figure shows the tips of two pins P_1 and P_2 kept in the upright positions. The parallax between P_1 and P_2 is removed by shifting the position of observer's eye sideways. As the farther pin P_1 is displaced towards the pin P_2 the relative shift (parallax) between their positions decreases as the position of eye is displaced sideways. The relative shift vanishes when the pin P_1 occupies the position P'_1, that is , when the tips of the two are just coincident. At this position there is no parallax between the tips of the two pins.

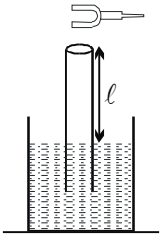


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7. DETERMINATION OF SPEED OF SOUND USING RESONANCE COLUMN

A tuning fork of known frequency (f) is held at the mouth of a long tube, which is dipped into water as shown in the figure. The length (l_1) of the air column in the tube is adjusted until it resonates with the tuning fork. The air temperature and humidity are noted. The length of the tube is adjusted again until a second resonance length (l_2) is found (provided the tube is long)

Then, $l_2 - l_1 = \lambda / 2$, provided l_1, l_2 are resonance lengths for adjacent resonances.



$\therefore\ \lambda = 2(l_2 - l_1)$, is the wavelength of sound.

Since the frequency f, is known; the velocity of sound in air at the temperature (θ) and humidity (h) is given by $C = f\lambda = 2(l_2 - l_1)f$

It is also possible to use a single measurement of the resonant length directly, but, then it has to be corrected *for* the "end effect":

λ(fundamental) = $4(l_1 + 0.3d)$, where d = diameter

Errors :

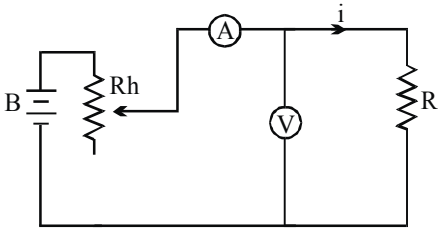
The major systematic errors introduced are due to end effects in (end correction) and also due to excessive humidity.

Random errors are given by $\frac{\delta C}{C} = \frac{\delta(l_2 - l_1)}{l_2 - l_1} = \frac{\delta l_2 + \delta l_1}{l_2 - l_1}$

8. VERIFICATION OF OHM'S LAW USING VOLTMETER AND AMMETER

A voltmeter (V) and an ammeter (A) are connected in a circuit along with a resistance R as shown in the figure, along with a battery B and a rheostat, Rh

Simultaneous readings of the current i and the potential drop V are taken by changing the resistance in the rheostat (Rh). A graph of *V* vs *i* is plotted and it is found to be linear (within errors). The magnitude of R is determined by either



(a) taking the ratio $\frac{V}{i}$ and then (b) fitting to a straight line: V = iR, and determining the slope R.

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Errors :

Systematic errors in this experiment arise from the current flowing through V (finite resistance of the voltmeter), the Joule heating effect in the circuit and the resistance of the connecting wires/ connections of the resistance. The effect of Joule heating may be minimized by switching on the circuit for a short while only, while the effect of finite resistance of the voltmeter can be overcome by using a high resistance instrument or a potentiometer. The lengths of connecting wires should be minimized as much as possible.

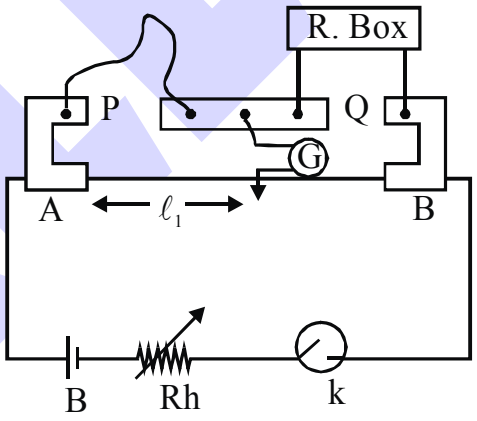
Error analysis :

The error in computing the ratio $R = \frac{V}{i}$ is given by $\left|\frac{\delta R}{R}\right| = \left|\frac{\delta V}{V}\right| + \left|\frac{\delta i}{i}\right|$

where δV and δi are of the order of the least counts of the instruments used.

9. SPECIFIC RESISTANCE OF THE MATERIAL OF A WIRE USING A METER BRIDGE:

A known length (l) of a wire is connected in one of the gaps (P) of a metre bridge, while a Resistance Box is inserted into the other gap (Q). The circuit is completed by using a battery (B), a Rheostat (Rh), a Key (K) and a galvanometer (G).The balance length (l) is found by closing key K and momentarily connecting the galvanometer until it gives zero deflection (null point). Then, $\frac{P}{Q} = \frac{l}{100 - l}$ ....(1)



using the expression for the meter bridge at balance. Here P represents the resistance of the wire while Q represents the resistance in the resistance box. The key K is open when the circuit is not in use.

The resistance of the wire, $P = \rho\frac{L}{\pi r^2} \Rightarrow \rho = \frac{\pi r^2}{L}P$ .....(2)

where r is the radius of wire and L is the length of the wire, r is measured using a screw gauge while L is measured with a scale.

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Errors :

The major systematic errors in this experiment are due to the heating effect, end corrections introduced due to shift of the zero of the scale at A and B, and stray resistances in P and Q, and errors due to non-uniformity of the meter bridge wire.

Error analysis :

End corrections can be estimated by including known resistances P_1 and Q_1 in the two ends and finding the null point:

$$\frac{P_1}{Q_1} = \frac{l_1 + \alpha}{100 - l_1 + \beta} \qquad \text{..... (2),}$$

where α and β are the end corrections.

When the resistance Q_1 is placed in the left gap and P_1 in the right gap,

$$\frac{Q_1}{P_1} = \frac{l_2 + \alpha}{100 - l_2 + \beta} \qquad \text{..... (3)}$$

which give two linear equation for finding α and β.

In order that α and β be measured accurately, P_1 and Q_1 should be as different from each other as possible. For the actual balance point, $\frac{P}{Q} = \frac{l + \alpha}{100 - l + \beta} = \frac{l_1'}{l_2'}$,

Errors due to non-uniformity of the meter bridge wire can be minimized by interchanging the resistances in the gaps P and Q. $\therefore \frac{\delta P}{P} = \left|\frac{\delta l_1'}{l_1'}\right| + \left|\frac{\delta l_2'}{l_2'}\right|$

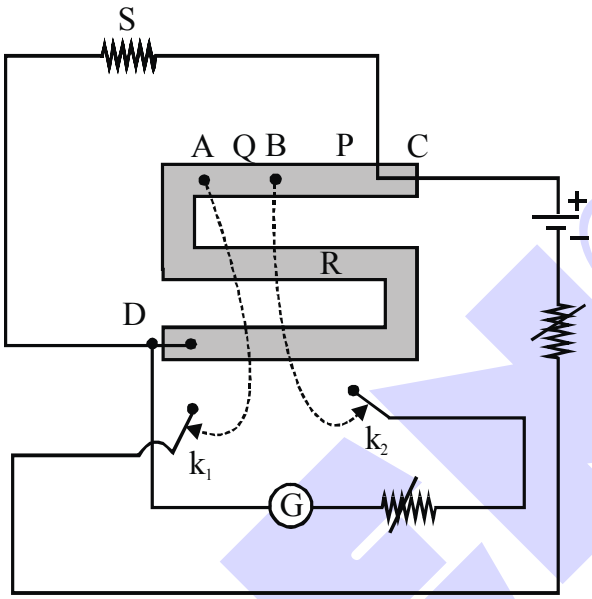
where, $\delta l'_1$ and $\delta l'_2$ are of the order of the least count of the scale.

The error is, therefore, minimum if $l'_1 = l'_2$ i.e. when the balance point is in the middle of the bridge. The error in ρ is $\frac{\delta\rho}{\rho} = \frac{2\delta r}{r} + \frac{\delta L}{L} + \frac{\delta P}{P}$

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10. MEASUREMENT OF UNKNOWN RESISTANCE USING A P.O. BOX

A P.O. Box can also be used to measure an unknown resistance. It is a Wheatstone Bridge with three arms P, Q and R; while the fourth arm(s) is the unknown resistance. P and Q are known as the ratio arms while R is known at the rheostat arm. At balance, the unknown resistance



$$S = \left(\frac{P}{Q}\right)R \qquad .....(1)$$

The ratio arms are first adjusted so that they carry 100 Ω each. The resistance in the rheostat arm is now adjusted so that the galvanometer deflection is in one direction, if $R = R_0$ (Ohm) and in the opposite direction when $R = R_0 + 1$ (ohm).

This implies that the unknown resistance, S lies between R_0 and $R_0 + 1$ (ohm). Now, the resistance in P and Q are made 100 Ω and 1000 Ω respectively, and the process is repeated.

Equation (1) is used to compute S.

The ratio P/Q is progressively made 1:10, and then 1 :100. The resistance S can be accurately measured.

Errors :

The major sources of error are the connecting wires, unclear resistance plugs, change in resistance due to Joule heating, and the insensitivity of the Wheatstone bridge.

These may be removed by using thick connecting wires, clean plugs, keeping the circuit on for very brief periods (to avoid Joule heating) and calculating the sensitivity.

In order that the sensitivity is maximum, the resistance in the arm P is close to the value of the resistance S.

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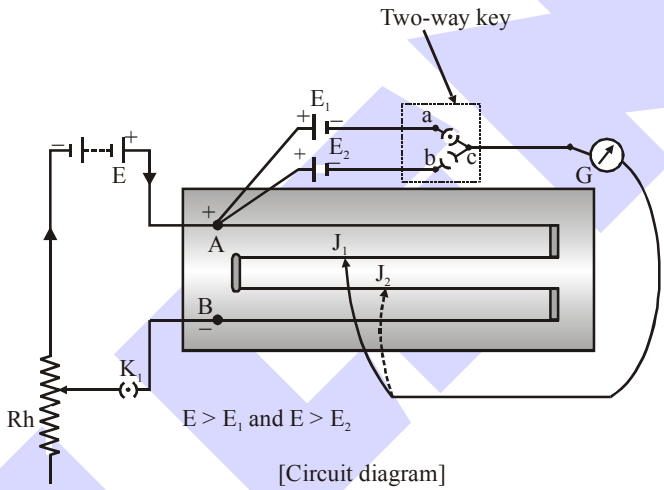
11. TO COMPARE THE EMF OF TWO GIVEN PRIMARY CELLS (DANIEL AND LECHLANCHE CELL) USING A POTENTIOMETER.

Apparatus required : A battery eliminator, a low resistance box about 20Ω, a one way key, two primary cells (Daniel and Lechlanche), one two way key, a resistance box plug type, a galvanometer, a potentiometer with a sliding jockey, connecting wires etc.

Formula required : $\frac{E_1}{E_2}=\frac{\ell_1}{\ell_2}$

where E_1 and E_2 are the emf's of the given primary cells and ℓ_1 and ℓ_2 are their respective balancing lengths on potentiometer wire.

Circuit diagram :



[Circuit diagram]

Procedure :

1. Arrange the circuit as shown in figure and plug key K_1.
2. In this circuit, by inserting the plug in the gap ac, cell E_1 comes in the secondary circuit. Introduce minimum resistance by Rh (rheostate) and then press the jockey at one end of the wire and note the direction of deflection in the galvanometer. Now repeat the same process with the jockey pressed near the other end of the wire and note the direction of deflection. The connections are correct if two deflections are in opposite directions.
3. If deflections are obtained in same direction, check connections and assure that $E > E_1$ and $E > E_2$.
4. After this, touch the jockey at different places on the wire and obtain a balance point, where on pressing the jockey, there is no deflection in the galvanometer note the length AJ_1 as ℓ_1 .
5. Introduce E_2 in the S circuit by transferring the plug ac to bc. Obtain the balancing point J_2 and note the length AJ_2 as ℓ_2.
6. Repeat the experiment by shifting the contact point of rheostate.

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Precautions :

1. The plugs should be introduced in the keys only when the observations are to be taken.
2. The positive poles of the battery and cells E_1 and E_2 should all be connected to the terminal at zero of the wires.
3. The emf of the battery should be greater than the emfs of either of the two cells.
4. The jockey should not be rubbed along the wire. It should touch the wire gently.
5. The current should not be passed for a long time so as to avoid any heating of the wires resulting into change of resistance.
6. The balancing poing should always be obtained at large distance and if possible it should be obtained at the last wire.

Source of error :

1. The battery (E) may not be fully charged.
2. By rubbing the jockey on the wire, it may not have uniform cross-section throughout its length.
3. The resistance of copper strips may not be zero.
4. Due to the current passed through the wire for a long time, its resistance may change.

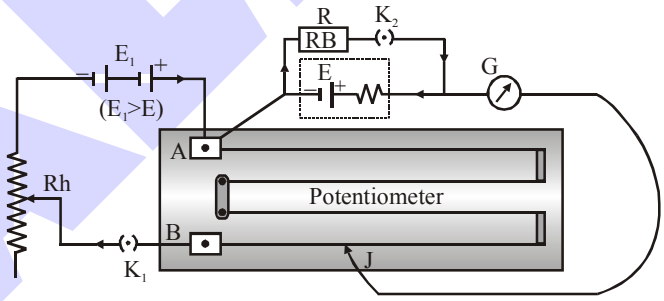
12. TO DETERMINE THE INTERNAL RESISTANCE OF A GIVEN PRIMARY CELL BY USING A POTENTIOMETER.

Apparatus required : Battery eliminator or cells in series, connecting wires, a rheostate, a galvanometer, a lechlanche cell, a resistance box, two one-way keys, a potentiometer with a sliding jockey etc.

Formula required : Internal resistance of a (primary) cell is $r = R\left(\frac{\ell_1}{\ell_2} - 1\right)$

where R–resistance used across the cell, ℓ_1 and ℓ_2 are the balancing lengths for the cell when it is in open and closed circuit respectively.

Circuit diagram :



[Circuit diagram]

Procedure :

1. Make the circuit as shown and ensure that the connections are tight and clean.
2. Keeping the key K_2 open, insert the plug in key K_1 and put the jockey near the end A and press it. Note the direction of deflection in galvanometer.
   Again note the direction of deflection by moving the jockey to the other end B and pressing.
   The deflection in these cases must be in opposite directions. If it is not so, then check the connections and adjust the rheostate so that the above condition is achieved.

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3. Keeping K_2 open and K_1 close, obtain the balancing length for the emf of cell and note it as ℓ_1.
4. Introduce some resistance (R) from the resistance box and close K_2 now and then obtain balancing length again and record it as ℓ_2.
5. Repeat the experiment for the different values of R and obtain different readings of ℓ_2.

Precautions :

1. Reading should be taken quickly as possible.
2. The cell should not be disturbed during the course of experiment.
3. The positive terminals of battery and cell should be connected at one point A.
4. Emf of the battery should be greater than the emf of lechanche cells.
5. All connections should be tight, neat and clean
6. The jockey should not be rubbed along the wire. It should touch the wire gently.

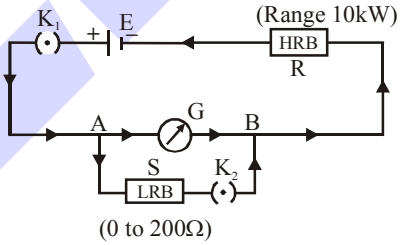
Source of error :

1. The battery (E) may not be fully charged.
2. By rubbing the jockey on the wire, it may not have uniform cross-section throughout its length.
3. The resistance of copper strips may not be zero.
4. Due to the current passed through the wire for a long time, its resistance may change.

13. TO DETERMINE THE RESISTANCE OF A GALVANOMETER BY HALF DEFLECTION METHOD AND FIGURE OF MERIT

Apparatus required : A galvanometer, a battery or accumulator, a low resistance box (LRB), a high resistance box (HRB), two one-way keys, connecting wires etc.

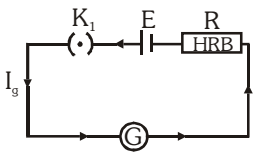
Circuit diagram :



[Circuit diagram]

(a) Resistance of galvanometer by half deflection method : When a high resistance R is applied in the circuit with K_1 closed and K_2 open, the galvanometer draws a current I_g and shows a deflection θ such that

$$I_g = \frac{E}{R+G} \qquad .....(i)$$



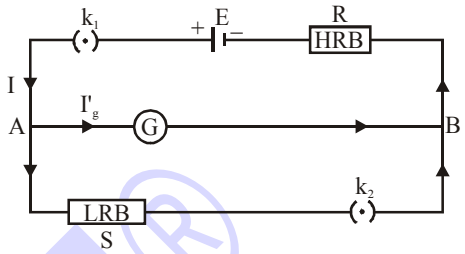
where E, is emf of the battery and G is resistance of the galvanometer.

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Now K_2 is closed. Adjust the resistance in LRB such that galvanometer deflection becomes equal to $\frac{\theta}{2}$. Now the galvanometer draw the current.

$$I'_g = \frac{IS}{G+S} = \frac{ES}{R(G+S)+GS} \qquad \left(\text{where } I = \frac{E}{R + \frac{GS}{G+S}}\right)$$

Also $I'_g = \frac{1}{2} I_g$



$\therefore \quad \frac{ES}{R(G+S)+GS} = \frac{1}{2}.\frac{E}{R+G}$

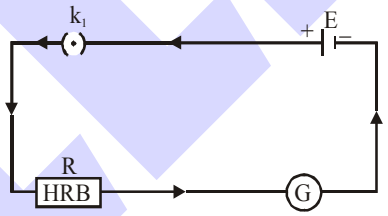
gives $G = \frac{RS}{R-S}$

Knowing R and S, G can be calculated.

Also if R > > S, S can be dropped in comparison to R and then $\boxed{G \simeq S}$.

(b) Figure of merit of galvanometer is defined as the current required per division of deflection in galvanometer. It is denoted by k. Figure merit $k = \frac{I}{\theta}$

The circuit diagram for determination of figure of merit (k) of a galvanometer is shown in the figure. When a high resistance R is introduced in the circuit through HRB, a small current I_g is drawn by it and it shows a deflection θ such that



$$I_g = k\theta = \frac{E}{R+G}$$

$\Rightarrow$ Figure of merit $k = \frac{1}{\theta}\left(\frac{E}{R+G}\right)$ ....(ii)

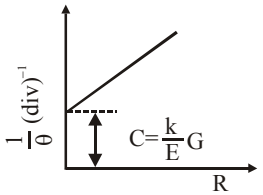
Maximum current measured by galvanometer or full scale deflection current for galvanometer

I_g = Number of division on one side of galvanometer scale × figure of merit

From equation (ii)

$$\frac{1}{\theta} = \frac{k}{E}R + \frac{k}{E}G \qquad .....(iii)$$

$\therefore$ Graph between $\frac{1}{\theta}$ and R is as shown.



[Graph between (1/θ) and R]

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Procedure :

1. In the circuit, introduce a high resistance R in HRB and then insert the plug in the key K_1. Adjust the value of R to get deflection θ in even number of divisions.
2. Now close the key K_2 also and adjust the shunt resistance S from low resistance box (LRB) to get a deflection exactly half of θ in galvanometer.
3. Repeat experiment for different values of R and θ.

Precautions :

1. All connections in the circuit should be neat, clean and tight.
2. All the plugs in HRB and LRB should be tight.
3. A very high resistance R from HRB should be introduced first and then key K_1 should be closed to avoid any over current damage in galvanometer.
4. The emf of the cell or battery should be constant.
5. The deflection in the galvanometer should be as large as possible and should be even number of divisions.

Sources of error

1. The plugs of HRB and LRB may not be clean.
2. The emf of the battery may not be constant.
3. The galvanometer divisions may not be uniform.

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